

Journal Supplementary I — Relating topics to labels and covariates: a visual annex

Companion to: *Beyond Accuracy* (cross-section relational visualisations)

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A.1 Scope and motivation

The journal main paper and its earlier supplements (A–H) defend the twelve-axis evaluation framework F-1..F-12 with figures that are either *per-axis scalars* (paired ARI, coherence, seed stability, etc.) or *per-topic intrinsic* (basis spectra, pairwise distances). What is left under-visualised is the **relational** question that motivates topic modelling in the first place: *do the topics actually relate to the labels and covariates we already have?* This supplement fills that gap with six figures, each chosen for a distinct cognitive question, with references to the underlying visualisation tradition.

B.2 F-7 viewed relationally: topic $\times$ class soft heatmap

This is the probabilistic generalisation of the confusion matrix. Each cell exposes *conditional label support*: a class-pure topic has one tall cell, a mixed topic spreads across columns. The top strip lets the reader gauge whether topic-weighted marginals match the unconditional class prior. Following the mosaic-display tradition of Friendly [1] and the single-cell-LDA precedent of TITAN’s HeatmapTopic.

C.3 F-7 again: per-topic class-distribution bars

The bars and the heatmap are duals; the heatmap optimises for finding outlier cells, the bars optimise for reading the within-topic distribution shape. Both ship.

D.4 Topic $\rightarrow$ class Sankey / alluvial flow

References: Sankey diagram tradition; Data-to-Viz Sankey-vs-alluvial distinction [4]; modern `d3-sankey` and `ggsankeyfier` implementations.

E.5 θ -PCA scatter coloured by topic / by label

Two-row design follows the `corner.py` scatterplot-matrix convention of Foreman-Mackey [5]; 2σ ellipses computed via the standard matplotlib gallery recipe [6].

F.6 Per-topic confidence ridge

The ridge plot is the appropriate idiom for “one continuous distribution per category”. We follow Wilke’s `ggridges` naming [2]; the precursor violin-plot family is Hintze and Nelson [3].

G.7 HIDSAG: topic $\times$ covariate tag

The continuous mineralogical measurements (Cu rec, Lime cons, mineral abundances per subset) live in the raw HDF5 source files of the figshare collection [8] rather than in the derived JSON exposed by our companion code repository. A future extension could plug those into the same topic–covariate frame using ridge plots per topic on the continuous variable, following the same idiom as Figure 5.

H.8 HIDSAG continuous-measurement ridges, corner plot, mosaic

After the initial release of this supplement, the HIDSAG continuous mineralogical / geometallurgical measurements were extracted from the per-sample `metadata.json` files in the figshare collection [8] (closing issue `CAOS_LDA_HSI_Paper#2`). A new derived layer at `data/derived/hidsag_topic_measurements/<subset>` joins the per-document θ matrix with the assay values; 427 records across the five subsets with 5–33 continuous variables per subset.

I.9 Why these nine figures

The nine figures answer nine distinct cognitive questions about the topic–label and topic–measurement relationships:

1. Heatmap (Fig. 1) — “which topic–class cells carry mass?”
2. Per-topic bars (Fig. 2) — “what is the within-topic class distribution shape?”
3. Sankey (Fig. 3) — “where does mass flow between topic and class?”
4. θ scatter (Fig. 4) — “do topics and labels align in the θ -PCA manifold?”
5. Confidence ridge (Fig. 5) — “how sharp is each topic’s posterior?”
6. HIDSAG covariate (Fig. 6) — “do topics carry information about the HIDSAG covariate tags?”
7. HIDSAG ridges (Fig. 7) — “does the dominant-topic assignment separate the continuous mineralogical assays?”
8. HIDSAG corner plot (Fig. 8) — “do topics occupy distinct manifolds in the GEOMET assay space?”
9. HIDSAG mosaic (Fig. 9) — “does the topic membership stochastically order each continuous variable?”

Each chosen idiom dominates its question on the cognitive-ergonomic trade-off between density (information per pixel) and decoding effort (steps the reader must take to extract a claim). See the research note in the repository `supplementary/journal/research-notes/topic-relational-viz.md` for the full literature survey that informed the selection.

References

- [1] M. Friendly, “Mosaic displays for multi-way contingency tables,” *Journal of the American Statistical Association*, vol. 89, no. 425, pp. 190–200, 1994.
- [2] C. O. Wilke, “ggridges: Ridgeline plots in ggplot2,” R package, <https://wilkelab.org/ggridges/>, 2017, companion blog post “Goodbye joyplots” (2017-09-15) explains the rename from joy plot.
- [3] J. L. Hintze and R. D. Nelson, “Violin plots: A box plot–density trace synergism,” *The American Statistician*, vol. 52, no. 2, pp. 181–184, 1998.

- [4] Data-to-Viz, “The sankey diagram,” <https://www.data-to-viz.com/graph/sankey.html>, 2024.
- [5] D. Foreman-Mackey, “corner.py: Scatterplot matrices in Python,” *Journal of Open Source Software*, vol. 1, no. 2, p. 24, 2016.
- [6] Matplotlib Developers, “Plot a confidence ellipse of a two-dimensional dataset,” Matplotlib gallery recipe, https://matplotlib.org/stable/gallery/statistics/confidence_ellipse.html, 2024.
- [7] A. Ehrenfeld, Á. F. Egaña, F. Santibañez-Leal, F. Garrido, M. Ojeda, B. Townley, and F. Navarro, “HIDSAG: Hyperspectral image database for supervised analysis in geometallurgy,” *Scientific Data*, vol. 10, no. 1, p. 164, 2023. [Online]. Available: <https://doi.org/10.1038/s41597-023-02061-x>
- [8] A. Ehrenfeld, “HIDSAG: Hyperspectral image database for supervised analysis in geometallurgy,” figshare Collection, <https://doi.org/10.6084/m9.figshare.c.5983921.v1>, 2023, data collection accompanying [7]; five thematic subsets (GEOMET, MINERAL1, MINERAL2, GEOCHEM, PORPHYRY). Reference implementation at <https://github.com/alges/hidsag>.
- [9] H. Hofmann, “Mosaic plots and their variants,” in *Handbook of Data Visualization*. Springer Berlin Heidelberg, 2008, pp. 617–642.

Topic $\times$ class soft heatmap — $P(\text{label} \mid \text{dominant topic})$ per scene. Top strip = label marginal weighted by topic prevalence.

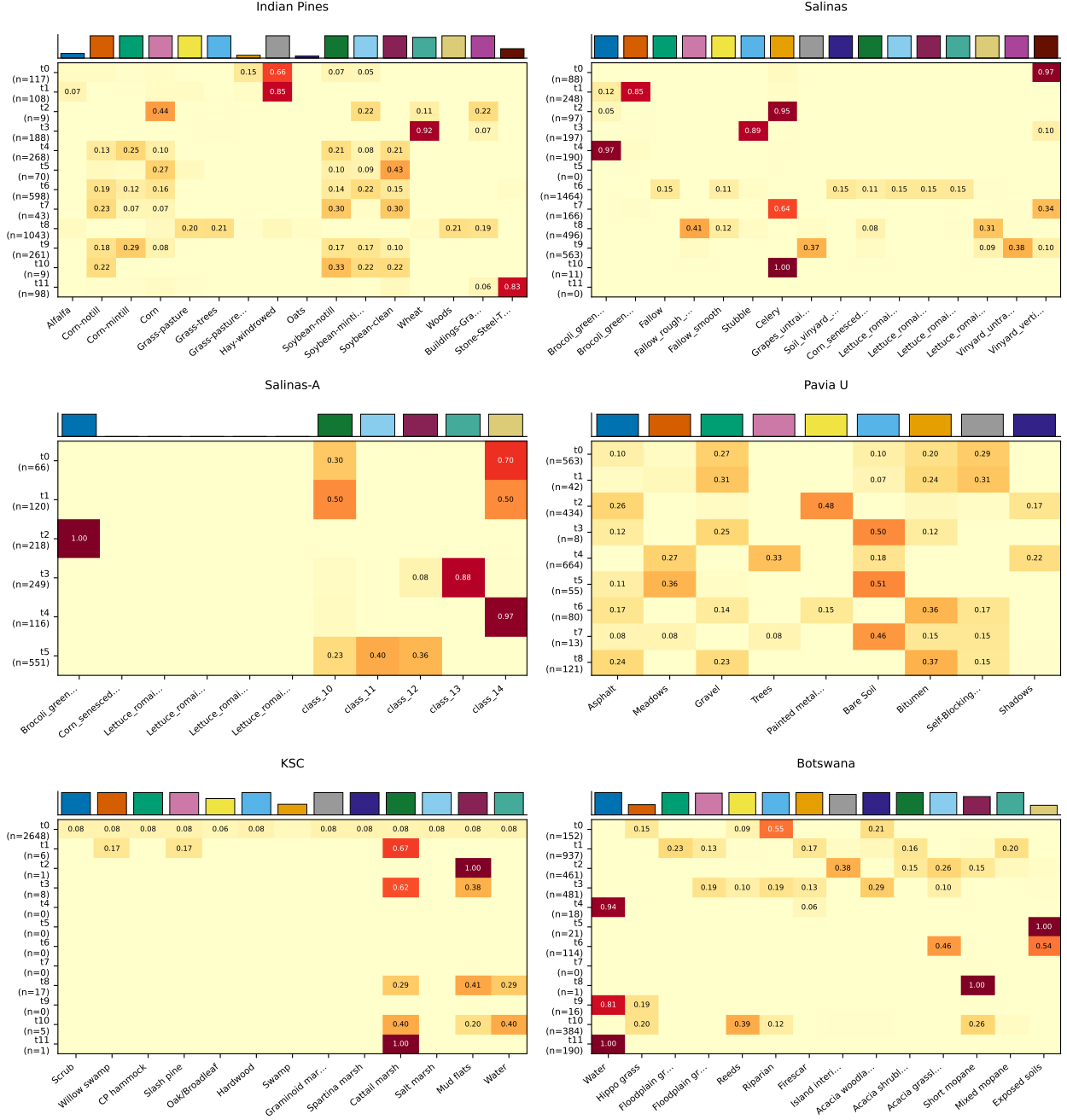


Figure 1: Topic $\times$ class soft heatmap per labelled scene. Rows = canonical topics (with document count); columns = ground-truth labels; cell intensity = $P(\ell \mid t)$, the conditional label distribution given dominant topic. Top strip = label marginal weighted by topic prevalence. Truncated class names appear on the x-axis.

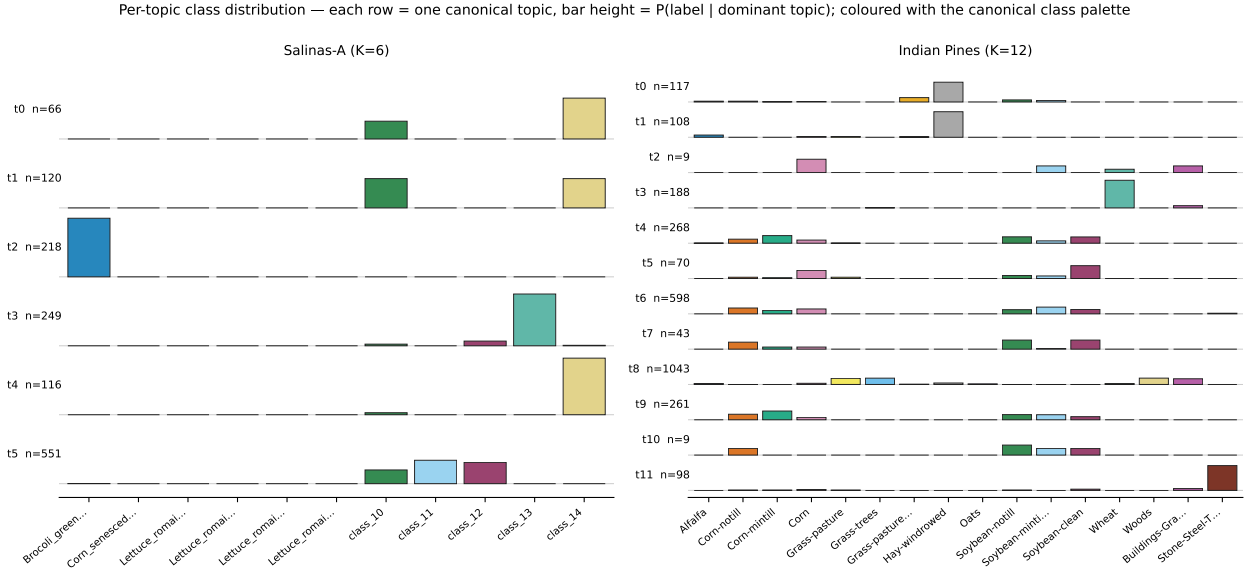


Figure 2: Per-topic class distribution. Each row inside a panel is one canonical topic; bar height $= P(\ell \mid t)$, coloured with the canonical class palette so the same class is visually identifiable across topics. A topic that resolves one or two classes shows as a single tall bar; a topic that mixes classes shows as a spread. The categorical analogue of the `ggribes` joy-plot pattern [2,3].

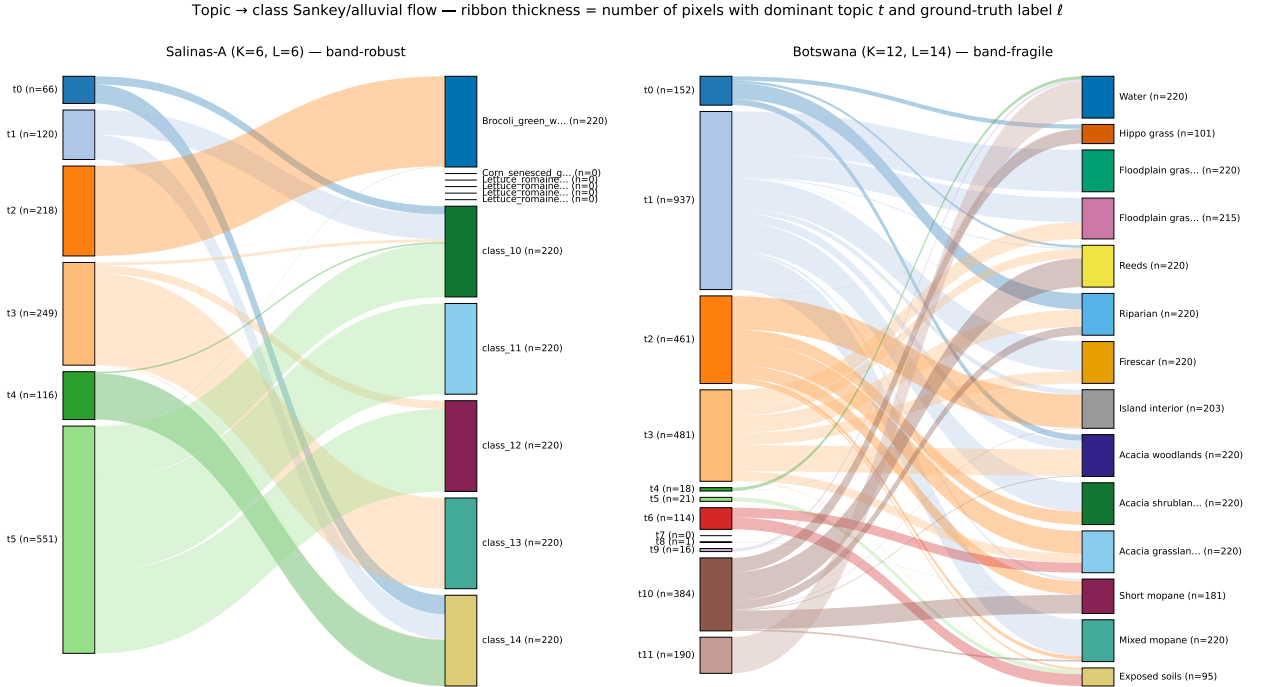


Figure 3: Topic $\rightarrow$ class Sankey/alluvial flow on Salinas-A (left) and Botswana (right). Left axis nodes = topics; right axis nodes = classes; ribbon width = pixel count. Salinas-A ($K=6$, $L=6$, band-robust) shows clean alignment with thin cross-flow. Botswana ($K=12$, $L=14$, band-fragile) shows the diffuse coupling that the F-5 paired ARI ≈ 0.01 predicts.

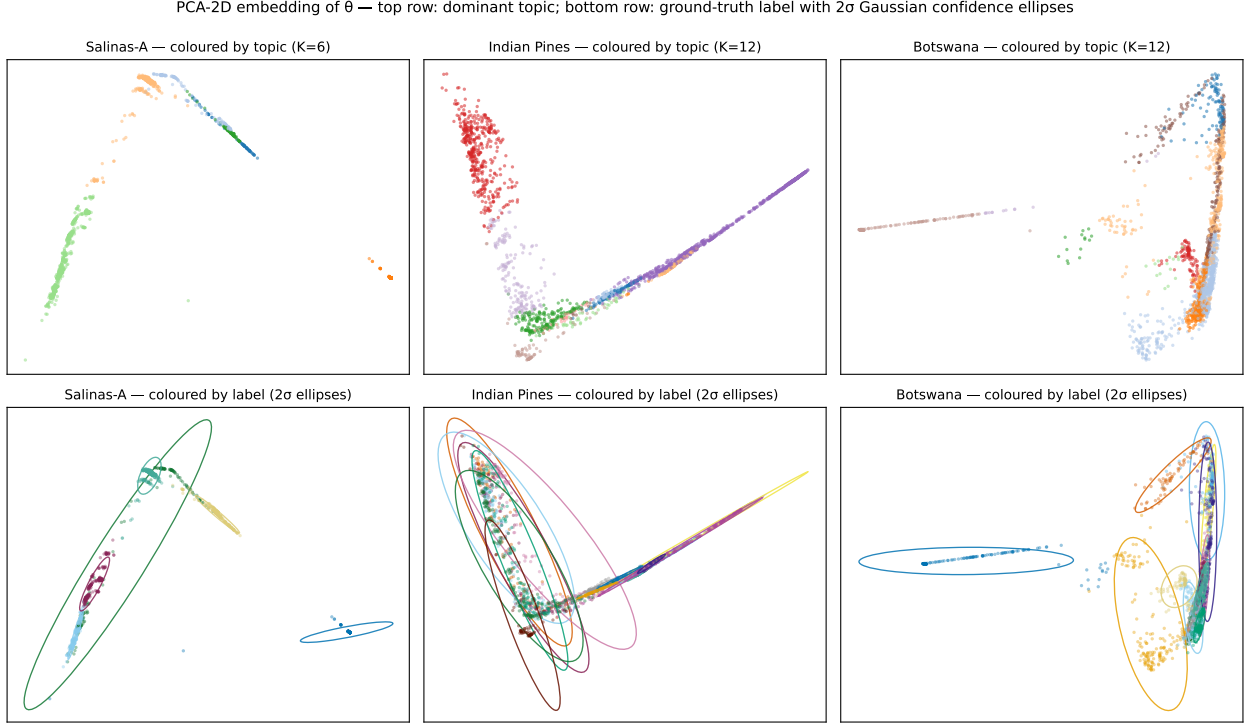


Figure 4: PCA-2D embedding of the per-document topic posterior θ_d on three scenes: top row coloured by dominant topic, bottom row coloured by ground-truth label with $2\text{-}\sigma$ Gaussian confidence ellipses per class. Salinas-A (band-robust): topic and label clusters align tightly. Indian Pines (partial): topic clusters visible but ellipses overlap. Botswana (band-fragile): topic clusters are visible but ground-truth-label ellipses span the whole embedding, exposing the topic–label decoupling.

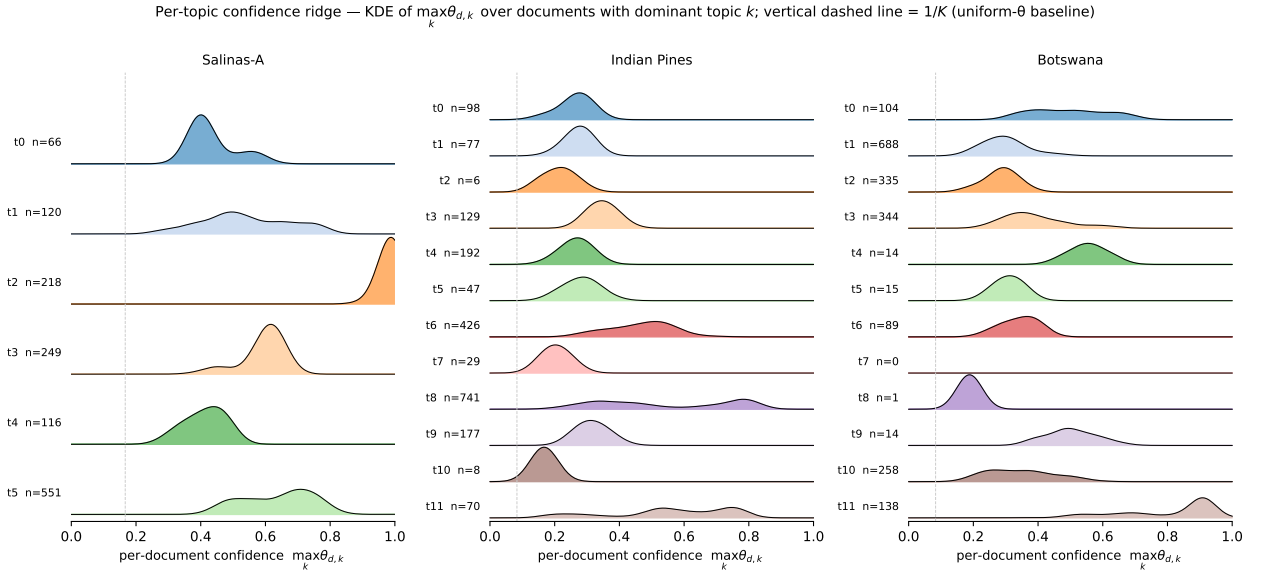


Figure 5: Per-topic KDE of the per-document confidence $\max_k \theta_{d,k}$ over documents with dominant topic k . Each topic gets one ridge curve on a shared x -axis $[0, 1]$; vertical dashed line at $1/K$ = uniform- θ baseline. Right-shifted curves = sharp topics; left-shifted = diffuse / collapsed topics.

HIDSAG topic $\times$ covariate-tag heatmaps — three subsets with non-degenerate tags. GEOMET and MINERAL2 omitted (single 'unknown' covariate, fall-back behaviour from the band-mask builder).

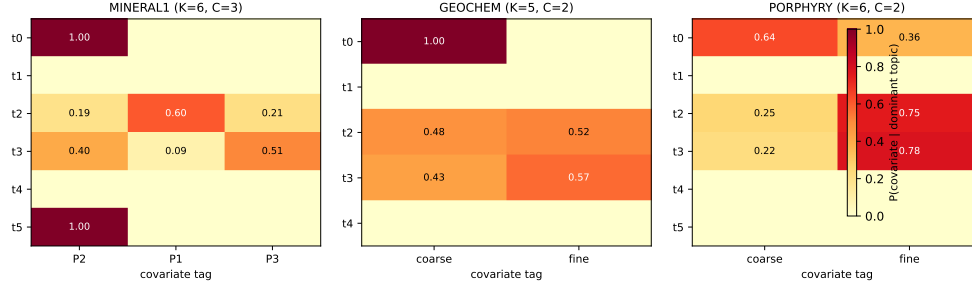


Figure 6: HIDSAG [7,8] topic $\times$ covariate-tag heatmap on three subsets with non-degenerate covariates (MINERAL1 with P1/P2/P3; GEOCHEM and PORPHYRY with coarse/fine). Cell = $P(\text{tag} \mid \text{dominant topic})$. GEOMET and MINERAL2 are omitted because their available covariates collapse to a single **unknown** value at the band-mask builder’s granularity (fall-back behaviour).

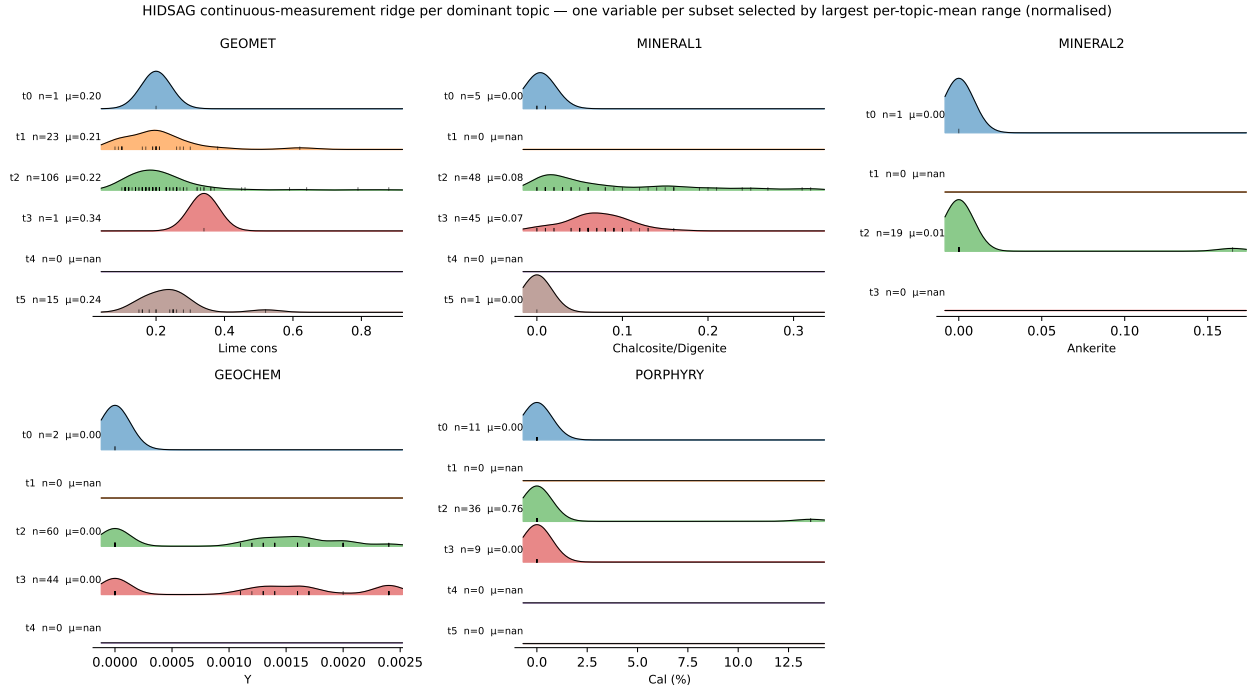


Figure 7: HIDSAG topic-conditional continuous-measurement ridges per subset. For each subset the variable with the largest normalised per-topic-mean range is selected; one KDE ridge per dominant topic on a shared x -axis (the variable). Per-row n and mean annotated. Rug marks at the actual observations. The most striking ridge is *PORPHYRY % biotite*: per-topic means range $1.1 \rightarrow 39.6$, indicating that the canonical-fit topic basis on PORPHYRY captures the porphyry-alteration mineralogy at a high level of fidelity.

GEOMET corner plot — 5 assays vs dominant topic (K=6; off-diagonal = 2σ Gaussian confidence ellipses; diagonal = KDE per topic). Foreman-Mackey 2016 *corner.py* idiom.

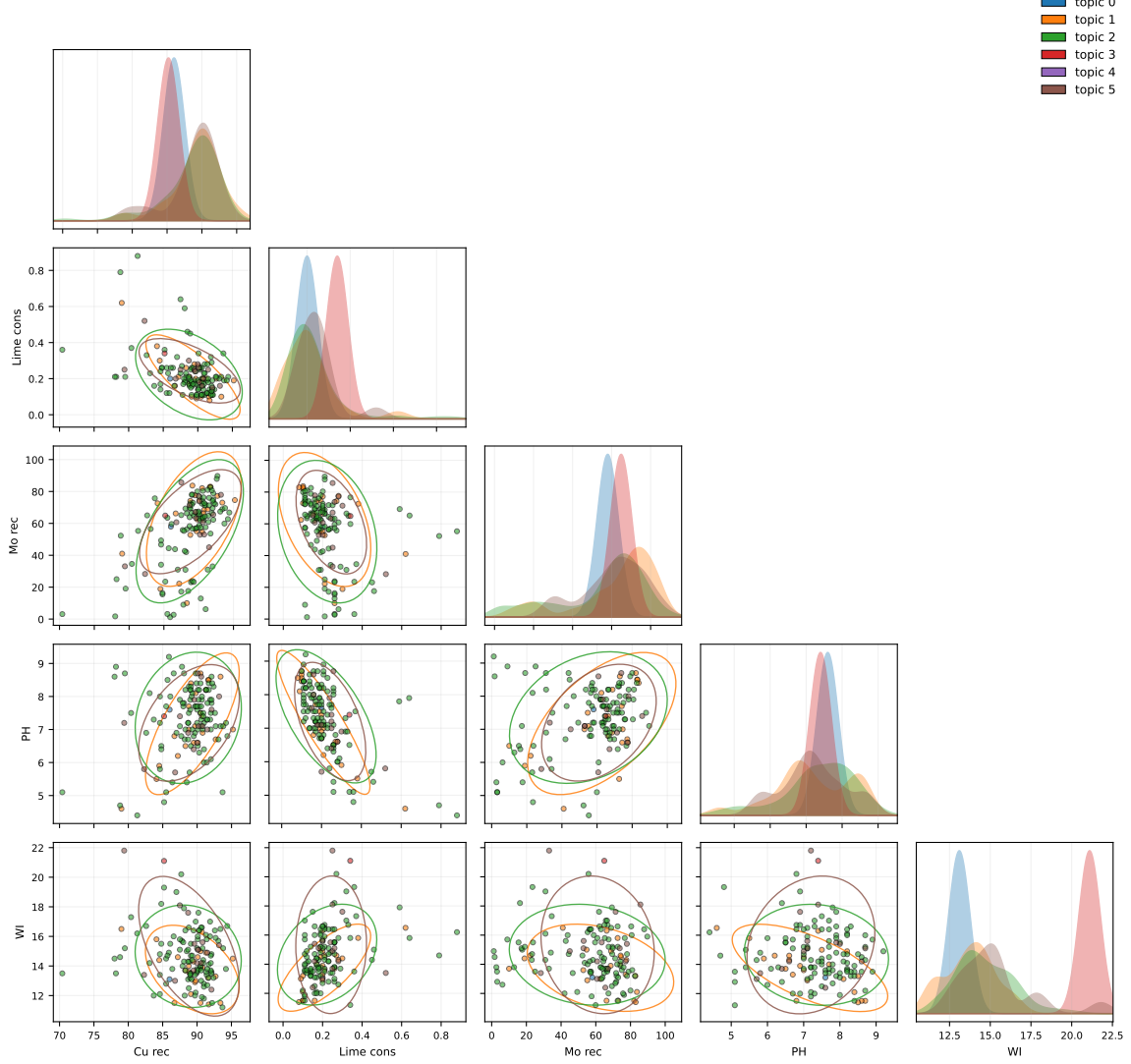


Figure 8: GEOMET corner-plot matrix coloured by dominant topic ($K = 6$). Diagonal: per-variable KDE per topic. Lower triangle: scatter of variable pairs coloured by topic with 2σ Gaussian confidence ellipses per topic. Idiom from Foreman-Mackey [5]; ellipses via the canonical matplotlib gallery recipe [6].

HIDSAG mosaic: $P(\text{quartile-bin} \mid \text{dominant topic})$ on a selected continuous variable per subset. Bins are scene-wide quartiles; rows are normalised so each topic sums to 1.

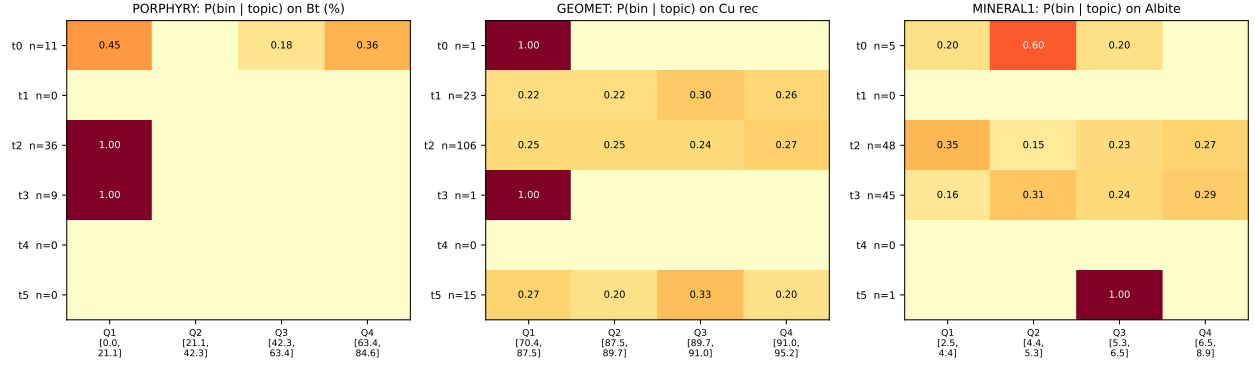


Figure 9: HIDSAG mosaic / fluctuation heatmap of $P(\text{quartile} \mid \text{dominant topic})$ on three subsets and the variable picked per subset (PORPHYRY: % biotite; GEOMET: Cu recovery; MINERAL1: albite). Bins are scene-wide quartiles; rows normalised. Captures non-monotone topic-vs-quartile patterns that the smooth ridge KDE can hide. Idiom: Hofmann [9]; Friendly [1].